Midterm Exam 1

55 minutes • 6 problems • 60 points

Name: _____ **Student ID:** _____

Instructions. This exam is closed-book and closed-notes. Calculators are not allowed. For each problem, show your reasoning, including the steps in calculations. You may quote results proved in class unless a problem explicitly asks for a proof.

- **Your work must be neat. If I can't read or find it, I can't grade it**.
- **Partial credit will only be given for correct and substantial work**.
- **Please put away your electronic devices**.

Honor statement: I agree to take this exam in accordance with the University Honor Code. My signature on this exam is a contractual statement asserting all work to be my own, that I have not collaborated in any way with my classmates, that I have not used outside resources, including the Internet, and that I have no knowledge that any of my classmates have broken University regulations on this exam.

Sign and date below.

_____ _____

Problem	**1**	**2**	**3**	**4**	**5**	**6**	**Total**
Points	10	8	14	8	10	10	**60**
Score							

Problem 1 **12 points**

(a) *[6 points]* Use the Euclidean algorithm to compute $\gcd(176, 68)$, and express the gcd as an integer linear combination of 176 and 68.

Solution. We run the Euclidean algorithm as follows:

$$\begin{aligned} 176 &= 68 \cdot 2 + 40 \\ 68 &= 40 \cdot 1 + 28 \\ 40 &= 28 \cdot 1 + 12 \\ 28 &= 12 \cdot 2 + 4 \\ 12 &= 4 \cdot 3 + 0. \end{aligned}$$

So $\gcd(176, 68) = 4$. Now read our calculations above backward:

$$\begin{aligned} 4 = 28 - 12 \cdot 2 &= 28 - (40 - 28 \cdot 1) \cdot 2 \\ &= 28 \cdot 3 - 40 \cdot 2 = (68 - 40 \cdot 1) \cdot 3 - 40 \cdot 2 \\ &= 68 \cdot 3 - 40 \cdot 5 = 68 \cdot 3 - (176 - 68 \cdot 2) \cdot 5 \\ &= 68 \cdot 13 - 176 \cdot 5. \end{aligned}$$

(b) *[4 points]* Find all integers $0 \leq x < 176$ satisfying the congruence

$$68x \equiv 12 \pmod{176}.$$

Solution. The linear combination that we found in Part (a) shows $68 \cdot 13 \equiv 4 \pmod{176}$. Multiplying both sides by 3 gives $68 \cdot (13 \cdot 3) \equiv 12 \pmod{176}$. So $x_0 = 13 \cdot 3 = 39$ is a solution.

Since $\gcd(176, 68) = 4$, there are exactly 4 solutions $0 \leq x < 176$ satisfying the given congruence. Let $m = 176/4 = 44$. Then the solutions are

$$x_0 = 39, x_0 + m = 39 + 44 = 83, x_0 + 2m = 83 + 44 = 127, x_0 + 3m = 127 + 44 = 171.$$

Problem 2 **8 points**

(a) *[4 points]* Compute the least nonnegative residue of

$$6^{203} \pmod{125}.$$

Solution. Since $\phi(125) = \phi(5^3) = 5^2(5-1) = 100$ and $\gcd(6, 125) = 1$, Euler's theorem gives $6^{100} \equiv 1 \pmod{125}$. Note that $203 = 100 \cdot 2 + 3$. Thus

$$6^{203} = \left(6^{100}\right)^2 \cdot 6^3 \equiv 6^3 \equiv 36 \cdot 6 \equiv 216 = 91 \pmod{125}.$$

(b) *[4 points]* Find the least nonnegative residue of

$$44! \pmod{47},$$

without expanding the factorial.

Solution. Because 47 is a prime, Wilson's theorem gives $46! \equiv -1 \equiv 46 \pmod{47}$. If $x \equiv 44! \pmod{47}$. Then

$$(-2)(-1)x \equiv 45 \cdot 46x \equiv 46! \equiv 46 \pmod{47}.$$

So $2x \equiv 46 \pmod{47}$. Since $\gcd(2, 47) = 1$, we may divide both sides by 2 to get $x \equiv 23 \pmod{47}$. In other words, $44! \equiv 23 \pmod{47}$.

Problem 3 **14 points**

(a) *[6 points]* Evaluate

$$\left(\frac{22}{59}\right).$$

Solution. We write

$$\left(\frac{22}{59}\right) = \left(\frac{2}{59}\right)\left(\frac{11}{59}\right).$$

Since $59 \equiv 3 \pmod 8$, we have

$$\left(\frac{2}{59}\right) = -1.$$

Since $11 \equiv 3 \pmod 4$, quadratic reciprocity gives

$$\left(\frac{11}{59}\right) = -\left(\frac{59}{11}\right) = -\left(\frac{4}{11}\right) = -1.$$

Hence

$$\left(\frac{22}{59}\right) = (-1)(-1) = 1.$$

(b) *[8 points]* Let p, q, r be distinct primes satisfying $p, q, r \equiv 3 \pmod 4$. Prove that exactly one or all three of the following congruences are solvable, with each congruence considered separately:

$$x^2 \equiv -qr \pmod p, \qquad y^2 \equiv -rp \pmod q, \qquad z^2 \equiv -pq \pmod r.$$

Hint: Compute the product

$$\left(\frac{-qr}{p}\right)\left(\frac{-rp}{q}\right)\left(\frac{-pq}{r}\right).$$

Solution. Since $p, q, r \equiv 3 \pmod 4$,

$$\begin{aligned}
\left(\frac{-qr}{p}\right) &= \left(\frac{-1}{p}\right)\left(\frac{q}{p}\right)\left(\frac{r}{p}\right) = -\left(\frac{q}{p}\right)\left(\frac{r}{p}\right), \\
\left(\frac{-rp}{q}\right) &= \left(\frac{-1}{q}\right)\left(\frac{r}{q}\right)\left(\frac{p}{q}\right) = -\left(\frac{r}{q}\right)\left(\frac{p}{q}\right), \\
\left(\frac{-pq}{r}\right) &= \left(\frac{-1}{r}\right)\left(\frac{p}{r}\right)\left(\frac{q}{r}\right) = -\left(\frac{p}{r}\right)\left(\frac{q}{r}\right).
\end{aligned}$$

Multiplying them out, we get

$$\left(\frac{-qr}{p}\right)\left(\frac{-rp}{q}\right)\left(\frac{-pq}{r}\right) = -\left(\frac{q}{p}\right)\left(\frac{r}{p}\right)\left(\frac{r}{q}\right)\left(\frac{p}{q}\right)\left(\frac{p}{r}\right)\left(\frac{q}{r}\right).$$

By quadratic reciprocity,

$$\begin{aligned}
\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) &= (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}} = -1, \\
\left(\frac{q}{r}\right)\left(\frac{r}{q}\right) &= (-1)^{\frac{q-1}{2}\cdot\frac{r-1}{2}} = -1, \\
\left(\frac{r}{p}\right)\left(\frac{p}{r}\right) &= (-1)^{\frac{r-1}{2}\cdot\frac{p-1}{2}} = -1.
\end{aligned}$$

So

$$\left(\frac{-qr}{p}\right)\left(\frac{-rp}{q}\right)\left(\frac{-pq}{r}\right) = -(-1)(-1)(-1) = 1.$$

Since each Legendre symbol takes values ± 1, the above identity can hold only when either exactly one of the Legendre symbol takes value 1 or all of the Legendre symbols take value 1. In other words, either exactly one or three of the considered congruences are solvable.

Problem 4 **10 points**

Suppose that p is an odd prime dividing $a^4 + 1$, where $a \in \mathbb{Z}$.

(a) *[3 points]* Prove that $p \equiv 1 \pmod 4$.

Solution. Since $p \mid a^4 + 1$, we have $a^4 + 1 \equiv 0 \pmod p$. Equivalently, $(a^2)^2 \equiv a^4 \equiv -1 \pmod p$. So -1 R p. This implies $p \equiv 1 \pmod 4$.

(b) *[5 points]* Use Fermat's little theorem to rule out $p \equiv 5 \pmod 8$, and deduce that $p \equiv 1 \pmod 8$.

Solution. Since $a^4 \equiv -1 \pmod p$, we have $p \nmid a$. Part (a) shows $p \equiv 1 \pmod 4$, so $p \equiv 1 \pmod 8$ or $p \equiv 5 \pmod 8$.

If $p \equiv 5 \pmod 8$, we can write $p = 8k + 5$ for some integer k. Then $a^4 \equiv -1 \pmod p$ implies

$$a^{p-1} = a^{8k+4} = \left(a^4\right)^{2k+1} \equiv (-1)^{2k+1} \equiv -1 \pmod p,$$

contradicting Fermat's little theorem $a^{p-1} \equiv 1 \pmod p$. So $p \equiv 1 \pmod 8$.

Problem 5 **10 points**

A positive integer n is called *squarefree* if n is not divisible by any square greater than 1. In other words, n is squarefree if $e_p(n) = 1$ for every $p \mid n$. For instance, $14 = 2 \cdot 7$ is squarefree, whereas $24 = 2^3 \cdot 3$ is not.

(a) *[8 points]* Prove that a squarefree positive integer n can be expressed as a sum of two integer squares if and only if the congruence

$$x^2 + 1 \equiv 0 \pmod{n}$$

has a solution.

Solution. If $x^2 + 1 \equiv 0 \pmod{n}$ has a solution, then $x^2 \equiv -1 \pmod{p}$ has a solution for every $p \mid n$. So $p = 2$ or -1 R p. Thus $p = 2$ or $p \equiv 1 \pmod{4}$. Hence, n has no prime factors congruent to 3 modulo 4. By the sum-of-two-squares theorem, n can be expressed as a sum of two integer squares.

Conversely, if $n = a^2 + b^2$, then $a^2 + b^2 \equiv 0 \pmod{n}$. If $p \mid n$ and $p \mid b$, then $p \mid a$. So $p^2 \mid a^2$ and $p^2 \mid b^2$, which implies $p^2 \mid n$, contradicting the hypothesis that n is squarefree. So $\gcd(b, n) = 1$. As a result, we can find a multiplicative inverse c of $b \pmod{n}$ such that $bc \equiv 1 \pmod{n}$. Multiplying $a^2 + b^2 \equiv 0 \pmod{n}$ by c^2, we get

$$(ac)^2 + 1 \equiv 0 \pmod{n}.$$

So $x = ac$ is a solution to $x^2 + 1 \equiv 0 \pmod{n}$.

(b) *[2 points]* Give a counterexample showing that the squarefree assumption in part (a) cannot be omitted.

Solution. Take $n = 2^2 = 4$ as an example. It can be expressed as a sum of two integer squares, but $x^2 \equiv -1 \pmod{4}$ has no solutions, because a square can only be congruent to 0 or 1 modulo 4.

One can also take $n = p^2$, where $p \equiv 3 \pmod{4}$ is any prime. The congruence $x^2 \equiv -1 \pmod{p^2}$ has no solutions, since otherwise $x^2 \equiv -1 \pmod{p}$ would have a solution, which is impossible because -1 N p.

Problem 6 **10 points**

Consider the ring $R = \mathbb{Z}[\sqrt{-10}]$.

(a) *[6 points]* Prove that $\gcd(2, \sqrt{-10}) = 1$ in R.

Solution. If $\alpha = x + y\sqrt{-10} \in \mathbb{Z}[\sqrt{-10}]$ is such that $\alpha \mid 2$ and $\alpha \mid \sqrt{-10}$, then $N(\alpha) \mid N(2) = 4$ and $N(\alpha) \mid N(\sqrt{-10}) = 10$. So $N(\alpha) \mid \gcd(4, 10) = 2$. Thus $N(\alpha) = 1$ or 2. But $N(\alpha) = x^2 + 10y^2 = 2$ has no solutions. So $N(\alpha) = 1$ and α is a unit. This shows that $\gcd(2, \sqrt{-10}) = 1$.

(b) *[4 points]* Show that there **do not** exist $\alpha, \beta \in R$ such that

$$2\alpha + \sqrt{-10}\,\beta = 1.$$

In particular, Bézout's lemma fails in R.

Solution. Write $\alpha = x + y\sqrt{-10}$ and $\beta = z + w\sqrt{-10}$. Then $2\alpha + \sqrt{-10}\,\beta = 1$ is equivalent to

$$(2x - 10w) + (2y + z)\sqrt{-10} = 1.$$

In particular, $2x - 10w = 1$, which is impossible because the left-hand side is even while the right-hand side is odd.

End of examination